

# Western Mathematics

## TOPIC REVISION QUESTIONS

# Mathematics Standard 2

## MS-F5 Annuities

**Section I – Multiple Choice Questions (1 Mark each)**

**Section II – Longer Questions (Marks indicated beside questions)**

## Section I

### 15 marks

1. Nate invests \$ 2000 at the end of each year into an account which pays compound interest at 6% pa and withdraws it after the last payment at the end of 4 years.

Which of the following would be the most help in finding the amount that he withdraws?

- (A) A formula for compound interest.
- (B) A formula for simple interest.
- (C) A table of future value interest factors.
- (D) A table of present value interest factors.
2. Use the table of future value interest factors below to find the future value of an annuity of \$3 000 per year for two years at 4% pa.

| Table of future value interest factors |                          |        |        |        |        |
|----------------------------------------|--------------------------|--------|--------|--------|--------|
|                                        | Interest rate per period |        |        |        |        |
| Period                                 | 1%                       | 2%     | 3%     | 4%     | 5%     |
| 1                                      | 1.0000                   | 1.0000 | 1.0000 | 1.0000 | 1.0000 |
| 2                                      | 2.0100                   | 2.0200 | 2.0300 | 2.0400 | 2.0500 |
| 3                                      | 3.0301                   | 3.0604 | 3.0909 | 3.1216 | 3.1525 |

- (A) \$3 509.00
- (B) \$6 120.00
- (C) \$6 240.00
- (D) \$6489.60

3. Georgie wants to make a single investment now which would produce the same outcome as an annuity of \$5000 per year for 3 years at 4% pa.

Use the table of Present value interest factors below to calculate the amount she needs to invest.

Present Value Interest Factors

| $r$         |   | 0.01    | 0.02    | 0.03    | 0.04    | 0.05    | 0.06    |
|-------------|---|---------|---------|---------|---------|---------|---------|
| Time period | 1 | 0.99010 | 0.98039 | 0.97087 | 0.96154 | 0.95238 | 0.94340 |
|             | 2 | 1.97040 | 1.94156 | 1.91347 | 1.88609 | 1.85941 | 1.83339 |
|             | 3 | 2.94099 | 2.88388 | 2.82861 | 2.77509 | 2.72325 | 2.67301 |
|             | 4 | 3.90197 | 3.80773 | 3.71710 | 3.62990 | 3.54595 | 3.46511 |
|             | 5 | 4.85343 | 4.71346 | 4.57971 | 4.45182 | 4.32948 | 4.21236 |
|             | 6 | 5.79548 | 5.60143 | 5.41719 | 5.24214 | 5.07569 | 4.91732 |
|             | 7 | 6.72819 | 6.47199 | 6.23028 | 6.00205 | 5.78637 | 5.58238 |
|             | 8 | 7.65168 | 7.32548 | 7.01969 | 6.73274 | 6.46321 | 6.20979 |

- (A) \$13 875.45
- (B) \$14 868.40
- (C) \$16 872.96
- (D) \$27 750.90
4. Joe invests \$400.00 at the end of each month into an account which pays 10.2% pa compounded monthly.

Use the table below to find the value of the investment after one and a half years

Future Value Interest Factors

| $r$         |    | 0.007   | 0.0075  | 0.008   | 0.0085  | 0.009   | 0.0095  |
|-------------|----|---------|---------|---------|---------|---------|---------|
| Time period | 16 | 16.8681 | 16.9323 | 16.9968 | 17.0616 | 17.1267 | 17.1921 |
|             | 17 | 17.9862 | 18.0593 | 18.1328 | 18.2066 | 18.2809 | 18.3555 |
|             | 18 | 19.1121 | 19.1947 | 19.2778 | 19.3614 | 19.4454 | 19.5298 |
|             | 19 | 20.2458 | 20.3387 | 20.4320 | 20.5260 | 20.6204 | 20.7154 |
|             | 20 | 21.3876 | 21.4912 | 21.5955 | 21.7004 | 21.8060 | 21.9122 |
|             | 21 | 22.5373 | 22.6524 | 22.7683 | 22.8849 | 23.0022 | 23.1203 |
|             | 22 | 23.6950 | 23.8223 | 23.9504 | 24.0794 | 24.2092 | 24.3400 |
|             | 23 | 24.8609 | 25.0010 | 25.1420 | 25.2841 | 25.4271 | 25.5712 |
|             | 24 | 26.0349 | 26.1885 | 26.3432 | 26.4990 | 26.6560 | 26.8141 |
|             | 25 | 27.2172 | 27.3849 | 27.5539 | 27.7242 | 27.8959 | 28.0689 |

- (A) \$6 824.64
- (B) \$7 282.64
- (C) \$7 711.12
- (D) \$7 744.56

Questions 5 and 6 refer to the following:

|                   |    | Future Value Interest Factors |         |         |         |         |         |         |
|-------------------|----|-------------------------------|---------|---------|---------|---------|---------|---------|
|                   |    | Interest Rate (r)             |         |         |         |         |         |         |
| Number of Periods | r  | 0.04                          | 0.05    | 0.06    | 0.07    | 0.08    | 0.09    | 0.1     |
|                   | 4  | 4.2465                        | 4.3101  | 4.3746  | 4.4399  | 4.5061  | 4.5731  | 4.6410  |
|                   | 5  | 5.4163                        | 5.5256  | 5.6371  | 5.7507  | 5.8666  | 5.9847  | 6.1051  |
|                   | 6  | 6.6330                        | 6.8019  | 6.9753  | 7.1533  | 7.3359  | 7.5233  | 7.7156  |
|                   | 7  | 7.8983                        | 8.1420  | 8.3938  | 8.6540  | 8.9228  | 9.2004  | 9.4872  |
|                   | 8  | 9.2142                        | 9.5491  | 9.8975  | 10.2598 | 10.6366 | 11.0285 | 11.4359 |
|                   | 9  | 10.5828                       | 11.0266 | 11.4913 | 11.9780 | 12.4876 | 13.0210 | 13.5795 |
|                   | 10 | 12.0061                       | 12.5779 | 13.1808 | 13.8164 | 14.4866 | 15.1929 | 15.9374 |
|                   | 11 | 13.4864                       | 14.2068 | 14.9716 | 15.7836 | 16.6455 | 17.5603 | 18.5312 |
|                   | 12 | 15.0258                       | 15.9171 | 16.8699 | 17.8885 | 18.9771 | 20.1407 | 21.3843 |
|                   | 13 | 16.6268                       | 17.7130 | 18.8821 | 20.1406 | 21.4953 | 22.9534 | 24.5227 |
|                   | 14 | 18.2919                       | 19.5986 | 21.0151 | 22.5505 | 24.2149 | 26.0192 | 27.9750 |
|                   | 15 | 20.0236                       | 21.5786 | 23.2760 | 25.1290 | 27.1521 | 29.3609 | 31.7725 |
|                   | 16 | 21.8245                       | 23.6575 | 25.6725 | 27.8881 | 30.3243 | 33.0034 | 35.9497 |

Richard plans to invest \$4500 at the end of every year into an account that pays 8% per annum compounded annually.

At the end of six years, he will withdraw the total balance, including the final instalment.

5. How much would he withdraw?
  - (A) \$26 399.70
  - (B) \$32 189.85
  - (C) \$33 011.55
  - (D) \$33 854.85
  
6. Richard decides to use another account into which he deposits half the amount above, but does so six-monthly rather than annually, with the interest also compounded six-monthly. How does the balance compare with that of the original plan?
  - (A) \$1 593 less than the original.
  - (B) \$796.50 more than the original.
  - (C) \$1 593 more than the original.
  - (D) The same as the original

Questions 7 and 8 refer to the following:

Jack and Mel take out a loan of \$42 000 to be repaid in monthly instalments over 5 years at 8.4% pa.

**Present Value Interest Factors**

|                   |     | Interest Rate ( $r$ ) |          |          |          |          |          |
|-------------------|-----|-----------------------|----------|----------|----------|----------|----------|
| Number of Periods | $r$ | 0.004                 | 0.005    | 0.006    | 0.007    | 0.008    | 0.009    |
|                   | 54  | 48.47937              | 47.22135 | 46.00820 | 44.83808 | 43.70922 | 42.61995 |
|                   | 55  | 49.28224              | 47.98145 | 46.72784 | 45.51944 | 44.35439 | 43.23087 |
|                   | 56  | 50.08191              | 48.73776 | 47.44318 | 46.19607 | 44.99443 | 43.83635 |
|                   | 57  | 50.87840              | 49.49031 | 48.15425 | 46.86799 | 45.62940 | 44.43642 |
|                   | 58  | 51.67171              | 50.23911 | 48.86109 | 47.53525 | 46.25932 | 45.03114 |
|                   | 59  | 52.46186              | 50.98419 | 49.56370 | 48.19786 | 46.88425 | 45.62055 |
|                   | 60  | 53.24887              | 51.72556 | 50.26213 | 48.85587 | 47.50421 | 46.20471 |
|                   | 61  | 54.03274              | 52.46324 | 50.95639 | 49.50931 | 48.11926 | 46.78366 |
|                   | 62  | 54.81348              | 53.19726 | 51.64651 | 50.15820 | 48.72942 | 47.35744 |
|                   | 63  | 55.59112              | 53.92762 | 52.33252 | 50.80258 | 49.33475 | 47.92611 |
|                   | 64  | 56.36566              | 54.65435 | 53.01443 | 51.44248 | 49.93526 | 48.48970 |
|                   | 65  | 57.13711              | 55.37746 | 53.69228 | 52.07794 | 50.53102 | 49.04827 |

7. How much is each instalment?
  - (A) \$824.23
  - (B) \$835.62
  - (C) \$848.33
  - (D) \$859.67
  
8. When Mel gets a promotion, they work out that they can afford to pay \$1 045 per month. How much extra could they borrow if the loan conditions were the same?
  - (A) \$9 054.38
  - (B) \$9 737.23
  - (C) \$10 523.93
  - (D) \$11 249.43



**Question 10** (4 marks)

Marcus borrows \$25 000 at 6% p.a. repaid in monthly instalments over 4.5 years.

- (a) Use the table of Present Value Interest factors to find his monthly repayment.

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| Present Value Interest Factors |          |          |          |          |          |          |          |          |
|--------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|
| Interest Rate (r)              |          |          |          |          |          |          |          |          |
|                                | 0.004    | 0.0045   | 0.005    | 0.0055   | 0.006    | 0.0065   | 0.007    | 0.0075   |
| 50                             | 45.23564 | 44.68459 | 44.14279 | 43.61006 | 43.08623 | 42.57113 | 42.06459 | 41.56645 |
| 51                             | 46.05143 | 45.47993 | 44.91820 | 44.36605 | 43.82329 | 43.28975 | 42.76524 | 42.24958 |
| 52                             | 46.86398 | 46.27170 | 45.68975 | 45.11790 | 44.55596 | 44.00373 | 43.46101 | 42.92762 |
| 53                             | 47.67329 | 47.05993 | 46.45746 | 45.86564 | 45.28425 | 44.71309 | 44.15195 | 43.60061 |
| 54                             | 48.47937 | 47.84463 | 47.22135 | 46.60929 | 46.00820 | 45.41787 | 44.83808 | 44.26860 |
| 55                             | 49.28224 | 48.62582 | 47.98145 | 47.34887 | 46.72784 | 46.11811 | 45.51944 | 44.93161 |
| 56                             | 50.08191 | 49.40350 | 48.73776 | 48.08440 | 47.44318 | 46.81382 | 46.19607 | 45.58969 |
| 57                             | 50.87840 | 50.17770 | 49.49031 | 48.81592 | 48.15425 | 47.50503 | 46.86799 | 46.24287 |
| 58                             | 51.67171 | 50.94843 | 50.23911 | 49.54343 | 48.86109 | 48.19179 | 47.53525 | 46.89118 |
| 59                             | 52.46186 | 51.71571 | 50.98419 | 50.26696 | 49.56370 | 48.87411 | 48.19786 | 47.53467 |
| 60                             | 53.24887 | 52.47956 | 51.72556 | 50.98653 | 50.26213 | 49.55202 | 48.85587 | 48.17337 |
| 61                             | 54.03274 | 53.23998 | 52.46324 | 51.70217 | 50.95639 | 50.22555 | 49.50931 | 48.80732 |
| 62                             | 54.81348 | 53.99699 | 53.19726 | 52.41389 | 51.64651 | 50.89474 | 50.15820 | 49.43654 |
| 63                             | 55.59112 | 54.75061 | 53.92762 | 53.12173 | 52.33252 | 51.55960 | 50.80258 | 50.06109 |
| 64                             | 56.36566 | 55.50086 | 54.65435 | 53.82568 | 53.01443 | 52.22017 | 51.44248 | 50.68098 |
| 65                             | 57.13711 | 56.24774 | 55.37746 | 54.52579 | 53.69228 | 52.87647 | 52.07794 | 51.29626 |

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- (b) How much does he pay in interest over the term of the loan?

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**Question 11** (4 marks)

Mr and Mrs Jones take out a loan for \$150 000 at an interest rate of 6% pa compounding monthly.

- (a) Use the table of present value interest factors to find the amount of the monthly repayment on the loan if it to be repaid in  $4\frac{1}{2}$  years. 2

| Present Value Interest Factors |          |          |          |          |          |          |          |          |
|--------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|
|                                | 0.004    | 0.0045   | 0.005    | 0.0055   | 0.006    | 0.0065   | 0.007    | 0.0075   |
| 48                             | 43.59425 | 43.08315 | 42.58032 | 42.08559 | 41.59882 | 41.11986 | 40.64856 | 40.18478 |
| 49                             | 44.41658 | 43.88567 | 43.36350 | 42.84992 | 42.34475 | 41.84785 | 41.35905 | 40.87820 |
| 50                             | 45.23564 | 44.68459 | 44.14279 | 43.61006 | 43.08623 | 42.57113 | 42.06459 | 41.56645 |
| 51                             | 46.05143 | 45.47993 | 44.91820 | 44.36605 | 43.82329 | 43.28975 | 42.76524 | 42.24958 |
| 52                             | 46.86398 | 46.27170 | 45.68975 | 45.11790 | 44.55596 | 44.00373 | 43.46101 | 42.92762 |
| 53                             | 47.67329 | 47.05993 | 46.45746 | 45.86564 | 45.28425 | 44.71309 | 44.15195 | 43.60061 |
| 54                             | 48.47937 | 47.84463 | 47.22135 | 46.60929 | 46.00820 | 45.41787 | 44.83808 | 44.26860 |
| 55                             | 49.28224 | 48.62582 | 47.98145 | 47.34887 | 46.72784 | 46.11811 | 45.51944 | 44.93161 |
| 56                             | 50.08191 | 49.40350 | 48.73776 | 48.08440 | 47.44318 | 46.81382 | 46.19607 | 45.58969 |
| 57                             | 50.87840 | 50.17770 | 49.49031 | 48.81592 | 48.15425 | 47.50503 | 46.86799 | 46.24287 |
| 58                             | 51.67171 | 50.94843 | 50.23911 | 49.54343 | 48.86109 | 48.19179 | 47.53525 | 46.89118 |
| 59                             | 52.46186 | 51.71571 | 50.98419 | 50.26696 | 49.56370 | 48.87411 | 48.19786 | 47.53467 |
| 60                             | 53.24887 | 52.47956 | 51.72556 | 50.98653 | 50.26213 | 49.55202 | 48.85587 | 48.17337 |
| 61                             | 54.03274 | 53.23998 | 52.46324 | 51.70217 | 50.95639 | 50.22555 | 49.50931 | 48.80732 |
| 62                             | 54.81348 | 53.99699 | 53.19726 | 52.41389 | 51.64651 | 50.89474 | 50.15820 | 49.43654 |

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- (b) How much additional interest would they pay, over the term of the loan, if they took it over 5 years rather than  $4\frac{1}{2}$  years? 2

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Question 12 (3 marks)

Gwen borrows \$53 000 to begin a new business. She has secured a 9% p.a. fixed interest loan. She makes monthly repayments and intends to complete the loan in 5 years.

(a) Use the table of Present Value Interest factors to find her monthly repayment.

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| Present Value Interest Factors |       |          |          |          |          |          |          |          |
|--------------------------------|-------|----------|----------|----------|----------|----------|----------|----------|
| Interest Rate (r)              |       |          |          |          |          |          |          |          |
| Number of Periods              | 0.004 | 0.0045   | 0.005    | 0.0055   | 0.006    | 0.0065   | 0.007    | 0.0075   |
|                                | 50    | 45.23564 | 44.68459 | 44.14279 | 43.61006 | 43.08623 | 42.57113 | 42.06459 |
|                                | 51    | 46.05143 | 45.47993 | 44.91820 | 44.36605 | 43.82329 | 43.28975 | 42.76524 |
|                                | 52    | 46.86398 | 46.27170 | 45.68975 | 45.11790 | 44.55596 | 44.00373 | 43.46101 |
|                                | 53    | 47.67329 | 47.05993 | 46.45746 | 45.86564 | 45.28425 | 44.71309 | 44.15195 |
|                                | 54    | 48.47937 | 47.84463 | 47.22135 | 46.60929 | 46.00820 | 45.41787 | 44.83808 |
|                                | 55    | 49.28224 | 48.62582 | 47.98145 | 47.34887 | 46.72784 | 46.11811 | 45.51944 |
|                                | 56    | 50.08191 | 49.40350 | 48.73776 | 48.08440 | 47.44318 | 46.81382 | 46.19607 |
|                                | 57    | 50.87840 | 50.17770 | 49.49031 | 48.81592 | 48.15425 | 47.50503 | 46.86799 |
|                                | 58    | 51.67171 | 50.94843 | 50.23911 | 49.54343 | 48.86109 | 48.19179 | 47.53525 |
|                                | 59    | 52.46186 | 51.71571 | 50.98419 | 50.26696 | 49.56370 | 48.87411 | 48.19786 |
|                                | 60    | 53.24887 | 52.47956 | 51.72556 | 50.98653 | 50.26213 | 49.55202 | 48.85587 |
|                                | 61    | 54.03274 | 53.23998 | 52.46324 | 51.70217 | 50.95639 | 50.22555 | 49.50931 |
|                                | 62    | 54.81348 | 53.99699 | 53.19726 | 52.41389 | 51.64651 | 50.89474 | 50.15820 |
|                                | 63    | 55.59112 | 54.75061 | 53.92762 | 53.12173 | 52.33252 | 51.55960 | 50.80258 |
|                                | 64    | 56.36566 | 55.50086 | 54.65435 | 53.82568 | 53.01443 | 52.22017 | 51.44248 |
|                                | 65    | 57.13711 | 56.24774 | 55.37746 | 54.52579 | 53.69228 | 52.87647 | 52.07794 |

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(b) Her financial advisor tells Gwen that the repayments should be less than 40% of her income.

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Calculate the income she needs to earn each month to cover this expectation.

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**Question 13** (3 marks)

The table below gives the future value interest factors for a range of loan terms and interest rates.

| Table of future value interest factors |                          |          |          |          |          |          |          |          |          |
|----------------------------------------|--------------------------|----------|----------|----------|----------|----------|----------|----------|----------|
|                                        | Interest Rate per Period |          |          |          |          |          |          |          |          |
| Period                                 | 0.35%                    | 0.40%    | 0.45%    | 0.50%    | 0.55%    | 0.60%    | 0.65%    | 0.70%    | 0.75%    |
| 115                                    | 141.2881                 | 145.6553 | 150.1958 | 154.9172 | 159.8275 | 164.9349 | 170.248  | 175.7761 | 181.5285 |
| 116                                    | 142.7826                 | 147.2379 | 151.8717 | 156.6918 | 161.7065 | 166.9245 | 172.3547 | 178.0065 | 183.89   |
| 117                                    | 144.2823                 | 148.8269 | 153.5551 | 158.4753 | 163.5959 | 168.926  | 174.475  | 180.2526 | 186.2692 |
| 118                                    | 145.7873                 | 150.4222 | 155.2461 | 160.2677 | 165.4957 | 170.9396 | 176.609  | 182.5143 | 188.6662 |
| 119                                    | 147.2976                 | 152.0239 | 156.9447 | 162.069  | 167.4059 | 172.9652 | 178.757  | 184.7919 | 191.0812 |
| 120                                    | 148.8131                 | 153.632  | 158.651  | 163.8793 | 169.3267 | 175.003  | 180.9189 | 187.0855 | 193.5143 |
| 121                                    | 150.334                  | 155.2465 | 160.3649 | 165.6987 | 171.258  | 177.053  | 183.0949 | 189.3951 | 195.9656 |
| 122                                    | 151.8601                 | 156.8675 | 162.0866 | 167.5272 | 173.1999 | 179.1153 | 185.285  | 191.7208 | 198.4354 |
| 123                                    | 153.3916                 | 158.4949 | 163.8159 | 169.3649 | 175.1525 | 181.19   | 187.4894 | 194.0629 | 200.9236 |
| 124                                    | 154.9285                 | 160.1289 | 165.5531 | 171.2117 | 177.1158 | 183.2772 | 189.7081 | 196.4213 | 203.4306 |
| 125                                    | 156.4708                 | 161.7694 | 167.2981 | 173.0678 | 179.09   | 185.3768 | 191.9412 | 198.7963 | 205.9563 |

- (a) Find the future value of an annuity of \$450 per month invested at 6.6% pa compounding monthly for 10 years. (Answer to the nearest dollar) **1**

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- (b) What is the minimum term required for an annuity of \$500 per month at 9% pa, compounding monthly to reach a value of \$100 000? **2**

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**Question 14** (3 marks)

The table shows present value interest factors for some monthly interest rates ( $r$ ) and loan terms in months ( $N$ ).

|                        | Monthly Interest Rate ( $r$ ) |         |         |         |         |         |         |
|------------------------|-------------------------------|---------|---------|---------|---------|---------|---------|
|                        | 0.0040                        | 0.0045  | 0.0050  | 0.0055  | 0.0060  | 0.0065  | 0.0070  |
| Term in months ( $N$ ) | 106                           | 86.2556 | 84.1540 | 82.1234 | 80.1612 | 78.2645 | 76.4310 |
|                        | 107                           | 86.9080 | 84.7725 | 82.7099 | 80.7172 | 78.7918 | 76.9309 |
|                        | 108                           | 87.5577 | 85.3883 | 83.2934 | 81.2702 | 79.3159 | 77.4277 |
|                        | 109                           | 88.2049 | 86.0013 | 83.8741 | 81.8202 | 79.8369 | 77.9212 |
|                        | 110                           | 88.8495 | 86.6115 | 84.4518 | 82.3672 | 80.3547 | 78.4115 |
|                        | 111                           | 89.4916 | 87.2190 | 85.0267 | 82.9112 | 80.8695 | 78.8987 |
|                        | 112                           | 90.1310 | 87.8238 | 85.5987 | 83.4522 | 81.3812 | 79.3827 |
|                        | 113                           | 90.7680 | 88.4259 | 86.1678 | 83.9903 | 81.8899 | 79.8636 |
|                        | 114                           | 91.4023 | 89.0253 | 86.7342 | 84.5254 | 82.3955 | 80.3413 |
|                        | 115                           | 92.0342 | 89.6220 | 87.2977 | 85.0576 | 82.8981 | 80.8160 |
|                        | 116                           | 92.6636 | 90.2160 | 87.8584 | 85.5868 | 83.3977 | 81.2877 |
|                        | 117                           | 93.2904 | 90.8074 | 88.4163 | 86.1132 | 83.8944 | 81.7562 |
|                        | 118                           | 93.9147 | 91.3961 | 88.9714 | 86.6367 | 84.3880 | 82.2218 |
|                        | 119                           | 94.5366 | 91.9822 | 89.5238 | 87.1573 | 84.8788 | 82.6844 |
|                        | 120                           | 95.1560 | 92.5656 | 90.0735 | 87.6751 | 85.3666 | 83.1439 |
|                        | 121                           | 95.7729 | 93.1465 | 90.6204 | 88.1901 | 85.8515 | 83.6005 |
|                        | 122                           | 96.3873 | 93.7247 | 91.1645 | 88.7022 | 86.3335 | 84.0542 |
|                        | 123                           | 96.9993 | 94.3004 | 91.7060 | 89.2115 | 86.8126 | 84.5049 |

Ruby borrows \$65 000 for home improvements. She repays the loan with monthly repayments over 10 years. She is charged 6% pa interest.

- (a) Calculate the amount of her monthly instalment. 1

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- (b) How much less interest would she pay if she took the loan over 9 years instead of 10 years? 2

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**End of Revision**

**Section I - Worked Solutions**

| <b>No</b> | <b>Working</b>                                                                                                                                                                                                                                        | <b>Answer</b> |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>1</b>  | This is an annuity and the future value is required, so a table of future value interest factors would be of most use. The compound interest formula could be used, but would be a much slower process.                                               | <b>C</b>      |
| <b>2</b>  | $FV = 2.0400 \times 3000 = \$6120$                                                                                                                                                                                                                    | <b>B</b>      |
| <b>3</b>  | $r = 0.04, n = 3 \Rightarrow \text{factor} = 2.77509$<br>Present Value = $\$5000 \times 2.77509 = \$13\,875.45$                                                                                                                                       | <b>A</b>      |
| <b>4</b>  | $r = 0.102 \div 12 = 0.0085, n = 1.5 \times 12 = 18 \Rightarrow \text{factor} = 19.3614$<br>Future Value = $\$400 \times 19.3614 = \$7\,744.56$                                                                                                       | <b>D</b>      |
| <b>5</b>  | $r = 0.08$ and $n = 6$<br>Interest factor = 7.3359<br>Balance = $7.3359 \times 4500 = \$33\,011.55$                                                                                                                                                   | <b>C</b>      |
| <b>6</b>  | $r = 0.04$ and $n = 12$<br>Interest factor = 15.0258<br>Balance = $15.0258 \times 2250 = \$33808.05$<br>Difference = $33808.05 - 33011.55 = \$796.50$ more than the original.                                                                         | <b>B</b>      |
| <b>7</b>  | $r = \frac{0.084}{12} = 0.007, n = 5 \times 12 = 60$<br>Factor = 48.85587<br>PV = $48.85587 \times \text{Instalment}$<br>$42\,000 = 48.85587 \times \text{Instalment}$<br>$\text{Instalment} = \frac{42000}{48.85587} = 859.67151951$<br>$= \$859.67$ | <b>D</b>      |
| <b>8</b>  | $r = \frac{0.084}{12} = 0.007, n = 5 \times 12 = 60$<br>Factor = 48.85587<br>PV = $48.85587 \times \text{Instalment}$<br>PV = $48.85587 \times 1045$<br>$= \$51054.38$<br>Extra = $51054.38 - 42000 = \$9\,054.38$                                    | <b>A</b>      |

| MS-F5 Annuities<br>Section II Solution |                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Question 9</b>                      |                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |
|                                        | $n = 7$<br>$r = 0.08$<br>Rate = 8.9228<br>$A = 8.9228 \times 2000 = \$17\,845.60$                                                                                                                                                                                                                                                                                                                                               | 2     |
| <b>Question 10</b>                     |                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |
| (a)                                    | $n = 12 \times 4.5 = 54$<br>$r = \frac{0.06}{12} = 0.005$<br>Rate = 47.22135<br>$25000 = M \times 47.22135$<br>$M = 25000 \div 47.22135 = 529.42154343$<br>= \$529.42                                                                                                                                                                                                                                                           | 2     |
| (b)                                    | Total repayed = $54 \times 529.42 = 28588.68$<br>Interest = $28588.68 - 25000$<br>= \$3 588.68                                                                                                                                                                                                                                                                                                                                  | 2     |
| <b>Question 11</b>                     |                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |
| (a)                                    | 6% pa = $6 \div 12 = 0.5\%$ per month ( $r = 0.005$ )<br>$4\frac{1}{2}$ years = $4\frac{1}{2} \times 12 = 54$ months ( $N = 54$ )<br>Interest factor = 47.22135<br>Amount of loan = Repayment $\times$ factor<br>150 000 = Repayment $\times$ 47.22135<br>Repayment = $\frac{150000}{47.22135} = \$3\,176.53$                                                                                                                   | 2     |
| (b)                                    | Over 54 months total amount paid = $54 \times 3176.53 = \$171\,532.62$<br>Interest paid = $171\,532.62 - 150\,000 = 21\,532.62$<br>Over 5 years (60 months) factor = 51.72556<br>Repayment = $\frac{150000}{51.72556} = \$2899.92$<br>Over 60 months total amount paid = $60 \times 2899.92 = \$173\,995.20$<br>Interest paid = $173\,995.20 - 150\,000 = 23\,995.20$<br>Extra interest = $23995.20 - 21532.62$<br>= \$2 462.58 | 2     |

| Question 12 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |          |          |          |          |          |          |          |          |    |          |          |          |          |          |          |          |          |    |          |          |          |          |          |          |          |          |   |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----|----------|----------|----------|----------|----------|----------|----------|----------|----|----------|----------|----------|----------|----------|----------|----------|----------|---|
| (a)         | $n = 60 \quad 12 \times 5$<br>$r = 0.0075 \quad 9\% \div 12$ <table><tr><td>59</td><td>52.70100</td><td>51.71971</td><td>50.98419</td><td>50.28090</td><td>49.59970</td><td>48.87411</td><td>48.19700</td><td>47.55407</td></tr><tr><td>60</td><td>53.24887</td><td>52.47956</td><td>51.72556</td><td>50.98653</td><td>50.26213</td><td>49.55202</td><td>48.85587</td><td>48.17337</td></tr><tr><td>61</td><td>54.02274</td><td>53.23008</td><td>52.46224</td><td>51.70217</td><td>50.95620</td><td>49.22555</td><td>48.50031</td><td>47.80722</td></tr></table><br>Factor = 48.17337<br>$53000 = M \times 48.17337...$<br>$M = 53000 \div 48.17337... = 1100.19$ | 59       | 52.70100 | 51.71971 | 50.98419 | 50.28090 | 49.59970 | 48.87411 | 48.19700 | 47.55407 | 60 | 53.24887 | 52.47956 | 51.72556 | 50.98653 | 50.26213 | 49.55202 | 48.85587 | 48.17337 | 61 | 54.02274 | 53.23008 | 52.46224 | 51.70217 | 50.95620 | 49.22555 | 48.50031 | 47.80722 | 2 |
| 59          | 52.70100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 51.71971 | 50.98419 | 50.28090 | 49.59970 | 48.87411 | 48.19700 | 47.55407 |          |          |    |          |          |          |          |          |          |          |          |    |          |          |          |          |          |          |          |          |   |
| 60          | 53.24887                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 52.47956 | 51.72556 | 50.98653 | 50.26213 | 49.55202 | 48.85587 | 48.17337 |          |          |    |          |          |          |          |          |          |          |          |    |          |          |          |          |          |          |          |          |   |
| 61          | 54.02274                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 53.23008 | 52.46224 | 51.70217 | 50.95620 | 49.22555 | 48.50031 | 47.80722 |          |          |    |          |          |          |          |          |          |          |          |    |          |          |          |          |          |          |          |          |   |
| (b)         | 40% of income = 1100<br>$0.4 \times I = 1100$<br>$I = \frac{1100}{0.4} = \$2\,750$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1        |          |          |          |          |          |          |          |          |    |          |          |          |          |          |          |          |          |    |          |          |          |          |          |          |          |          |   |
| Question 13 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |          |          |          |          |          |          |          |          |    |          |          |          |          |          |          |          |          |    |          |          |          |          |          |          |          |          |   |
| (a)         | 6.6% pa $\% = \frac{6.6}{12} = 0.55\%$ per month.<br><br>From table interest factor for 0.55% for 12 years<br>(120 months) is 169.3267.<br><br>Future value = $450 \times 169.3267 = \$76\,197$ (nearest dollar)                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1        |          |          |          |          |          |          |          |          |    |          |          |          |          |          |          |          |          |    |          |          |          |          |          |          |          |          |   |
| (b)         | 9.0% pa $\% = \frac{9.0}{12} = 0.75\%$ per month.<br><br>Let $I$ be the required interest factor<br>Then $500 \times I = \$100\,000$<br>$I = \frac{100000}{500} = 200$<br><br>From table, in the column for 0.75% the first interest factor to exceed 200 is for 123 months (200.9236),<br><br>So the minimum term is 10 years and 3 months                                                                                                                                                                                                                                                                                                                       | 2        |          |          |          |          |          |          |          |          |    |          |          |          |          |          |          |          |          |    |          |          |          |          |          |          |          |          |   |

| Question 14 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| (a)         | <p>6% p.a. interest = 0.5% per month.</p> <p>So <math>r = 0.0050</math></p> <p>Ten years is 120 months, so <math>N = 120</math></p> <p>From table the factor is 90.0735.</p> <p>So <math>90.0735 \times \text{Monthly repayment} = 65000</math></p> $\text{Monthly repayment} = \frac{65000}{90.0735}$ $= \$721.63$                                                                                                                                                                                      | 1 |
| (b)         | <p>Total repaid (ten years) = <math>721.63 \times 120 = \\$86\,595.60</math></p> <p>Nine years is 108 months, so <math>N = 108</math></p> <p>From table the factor is 83.2934.</p> <p>So <math>83.2934 \times \text{Monthly repayment} = 65000</math></p> $\text{Monthly repayment} = \frac{65000}{83.2934}$ $= \$780.37$ <p>Total repaid (nine years) = <math>780.37 \times 108 = \\$84\,279.96</math></p> <p>Amount of interest saved = <math>\\$86\,595.60 - \\$84\,279.96</math></p> $= \$2\,315.64$ | 2 |